5′-AMP-activated protein kinase regulates skeletal muscle glycogen content and ergogenics

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ABSTRACT **5′-AMP-activated protein kinase (AMPK) activity is increased during exercise in an intensity- and glycogen-dependent manner. We previously reported that a mutation in the AMPKγ3 subunit (*Prkag3*^{225Q}) increases AMPK activity and skeletal muscle glycogen content. Transfection experiments revealed the R225Q mutation is associated with high basal AMPK activity and diminished AMP dependence. Thus, the R225Q mutation can be considered a loss-of-function mutation that abolished allosteric regulation by AMP/ATP, causing increased basal AMPK activity. We used AMPKγ3 transgenic (*Tg-Prkag3*^{225Q}) and knockout (*Prkag3*$^{-/-}$) mice to determine the relationship between AMPK activity, glycogen content, and ergogenics (ability to perform work) in isolated extensor digitorum longus skeletal muscle after contractions induced by electrical stimulation. Contraction-induced AMPK activity was inversely coupled to glycogen content in wild-type and *Tg-Prkag3*^{225Q} mice, but not in *Prkag3*$^{-/-}$ mice, highlighting a partial feedback control of glycogen on contraction-induced AMPK activity in the presence of a functional AMPKγ3 isoform. Skeletal muscle glycogen content was positively correlated to work performance, regardless of genotype. Thus, chronic activation of AMPK by the *Prkag3*^{225Q} mutation directly influences skeletal muscle ergogenics by enhancing glycogen content. In conclusion, functional studies of the AMPKγ3 isoform further support the close connection between glycogen content and exercise performance in skeletal muscle.—Barnes, B. R., Glund, S., Long, Y. C., Hjälm, G., Andersson, L., Zierath, J. R. 5′-AMP-activated protein kinase regulates skeletal muscle glycogen content and ergogenics. *FASEB J.* 19, 773–779 (2005)**

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Over the past 40 years, exercise physiologists have appreciated the finding that an increase in dietary carbohydrate intake in the days before competition increases muscle glycogen levels and enhances exercise performance in power and endurance events. This increase in muscle glycogen content above that of the untrained state is recognized as glycogen supercompensation. Glycogen supercompensation can occur within the first few hours after cessation of work (1), and thus a rapid regulatory event rather than protein synthesis is required. However, the biological mechanism for glycogen supercompensation after exercise is incompletely resolved.

Exercise induces an insulin-independent effect on glucose uptake, which partly accounts for rapid glycogen resynthesis in the recovery phase (2, 3). This "exercise effect" on glucose transport persists until muscle glycogen levels are restored (2, 4), indicating a feedback regulation of glycogen on glucose transport. Although the acute effects of exercise on substrate utilization has been a topic of great interest, the signaling pathways by which muscle contraction promotes glycogen resynthesis are incompletely described. Historically, glycogen synthase has been highlighted as a major regulatory step in glycogen resynthesis in response to exercise. Functional studies in transgenic mice whereby glycogen synthase has been overexpressed in skeletal muscle provide evidence that the level of glycogen synthase activity has a positive impact on glycogen content (5). Moreover, under physiological conditions such as strenuous exercise, the rate of glycogen repletion is positively correlated with glycogen synthase activity (6). However, exercise-induced signaling pathways to glycogen synthase are unknown. Recent evidence places the 5′AMP-activated protein kinase (AMPK) signaling cascade upstream of glycogen synthase (7, 8). Thus, AMPK may be involved in post-exercise glycogen resynthesis.

AMPK is a heterotrimeric protein composed of one catalytic (α) and two noncatalytic (β and γ) subunits that, when activated, stimulates catabolic processes to produce ATP (9). Exercise increases AMPK activity in skeletal muscle (10), with greatest effects occurring

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when glycogen levels are low (11, 12). Thus, AMPK activation appears to be intimately linked to the preexercise fuel status of skeletal muscle irrespective of changes in energy status (11, 12). The molecular link between AMPK and glycogen content has been directly established through genetic studies. Overexpression of a kinase dead AMPKα2 isoform or ablation of the γ3 subunit is associated with impaired glycogen resynthesis in skeletal muscle (13, 14). In contrast, naturally occurring mutations in the γ2 and γ3 subunit isoforms increase glycogen storage in human heart and pig skeletal muscle, respectively (15). The increase in muscle glycogen that occurs as a consequence of mutations in the γ3 isoform may enhance glycogen storage by shifting the metabolic fate of glucose from oxidation to storage, but this is unexplored.

The relationship between AMPK activation, glycogen content, glucose utilization, and muscle ergogenics (the ability to perform work) remains unclear. We previously reported that the dominant missense mutation (R225Q) identified in pig PRKAG3 encoding the muscle-specific γ3 isoform causes a marked increase in glycogen content (16). Transfection experiments revealed the R225Q mutation is associated with high basal AMPK activity and diminished AMP dependence (13). Thus, the R225Q mutation can be considered a loss-of-function mutation that abolishes allosteric regulation by AMP/ATP, causing increased basal AMPK activity. Here we provide evidence that glycogen supercompensation requires expression of the AMPKγ3 isoform in the fed state. Moreover R225Q mutation promotes a shift in glucose metabolism from oxidation to storage in response to exercise. The elevated muscle glycogen provides an ergogenic benefit by increasing performance directly in working muscle.

MATERIALS AND METHODS

Animals and isolated skeletal muscle procedure

Three animal models were used: transgenic mice with skeletal muscle-specific expression, a mutant form of the AMPKγ3 subunit (*Tg-Prkag3*^{225Q}); knockout mice, whereby the *Prkag3* gene has been ablated (*Prkag3*$^{-/-}$); and wild-type littermates. The general metabolic characteristics of these animal models have been described (13). Mice were maintained in a temperature- and light-controlled environment and were cared for in accordance with regulations for the protection of laboratory animals. The regional animal ethical committee approved all experimental procedures. Animals had free access to water and standard rodent chow and were studied under fed or overnight fasted conditions. Mice were anesthetized with Avertin [2,2,2-tribromo ethanol 99+% and tertiary amyl alcohol (0.015–0.017 mL/g body weight)] and extensor digitorum longus (EDL) muscles were isolated. Muscles were preincubated at 30°C in Krebs-Henseleit bicarbonate buffer containing 5 mmol/L glucose and 15 mmol/L mannitol for 60 min. Media were continuously gassed with 95% O_2/5% CO_2 throughout the incubation procedures.

In vitro muscle contraction

EDL muscles were mounted in a temperature-controlled (30°C) incubation chamber, positioned between two platinum electrodes and immersed in media (13, 17). The proximal tendon was connected to an isometric force transducer (Harvard Apparatus, South Natick, MA, USA). One muscle from each animal was stimulated at 100 Hz (0.2 ms pulse duration, 20 V) at a rate of one 0.2 s contraction every 2 s for 10 min (18). The contralateral muscle was incubated under resting (nonstimulated) conditions. Incubations were terminated by immediate freezing of the muscle in liquid nitrogen. Muscles were stored at −80°C.

Total AMPK activity

Kinase assays were performed as described (19). pan-AMPK antiserum raised against bacterially expressed α1, α2, β1, β2, and γ1 protein (generous gift from David Carling, Imperial College, Hammersmith Hospital, London, UK) was used to precipitate total AMPK protein. Five μL pan-AMPK antiserum prebound to 20 μL protein A-Sepharose (50% slurry; Amersham Biosciences, Uppsala, Sweden) was used to immunoprecipitate the kinase from skeletal muscle lysate (200 μg protein) at 4°C for 3 h. Total AMPK activity was determined in the washed immune complex using SAMS peptide as substrate. Complexes were incubated in assay buffer for 45 min at 30°C. Incorporation of [^{32}P]-ATP into the peptide was measured by liquid scintillation counting of sample aliquots spotted on P81 paper (Whatman International, Maidstone, UK).

Glycogen content

Glycogen content was determined fluorometrically on HCl extracts (20).

Skeletal muscle fatigue

During in vitro electrical stimulation, the force of contraction generated by the muscle was measured, quantified by an isometric force transducer (Harvard Apparatus), and recorded with an oscilloscope. Peak force (initial contraction) generation was recorded for each muscle. Thereafter, muscle fatigue was determined by measuring the time expired before force generation equaled 50% of the peak force.

Glucose oxidation

Muscles were incubated at 30°C for 60 min in incubation media described above, supplemented with [^{14}C]-glucose (0.2 mCi/mL), as described (21). Thereafter, 0.2 mL solvent was injected into the center well of the incubation flask and 0.5 mL of 15% perchloric acid was injected into the incubation media. Glucose oxidation was assessed by collection of liberated [^{14}C]O_2.

Statistical analysis

A difference between groups was determined by ANOVA with multiple comparisons for AMPK activity. Student's t test was used for all other analyses. Significance was accepted at $P<0.05$. Relationships were determined by linear correlation and regression analysis.

TABLE 1. *Total AMPK activity in glycolytic skeletal muscle*[a]

	Fed		Fasted	
	Basal	Contraction	Basal	Contraction
Wild-type	4.47 ± 0.48	14.71 ± 1.35†	5.69 ± 0.60	16.58 ± 1.39†
Tg-Prkag3^{225Q}	3.61 ± 0.34	9.05 ± 0.98†*	5.73 ± 0.67	19.71 ± 1.70†
Prkag3$^{-/-}$	3.74 ± 0.11	7.34 ± 0.65†*	2.74 ± 0.73*	16.91 ± 1.70†

[a] Total AMPK activity was measured in EDL skeletal muscle in *wild-type, Tg-Prkag3*^{225Q}, and *Prkag3*$^{-/-}$ mice under fed or fasted conditions before or after *in vitro* muscle contraction. Values are expressed as pmol × mg^{-1} × min^{-1} (n=4–9). * $P < 0.05$ vs. *wild-type* of same condition. † $P < 0.05$ vs. basal of same genotype.

RESULTS

AMPK activity

AMPK activity was measured in EDL skeletal muscle extracts from wild-type, *Tg-Prkag3*^{225Q}, and *Prkag3*$^{-/-}$ mice after immunoprecipitation with a pan-AMPK antibody (**Table 1**). Under fed conditions, basal AMPK activity was similar among genotypes. In vitro contraction of isolated EDL muscle from fed and fasted wild-type mice increased AMPK activity by 220% (P<0.05) and 190% (P<0.05), respectively. Contraction-induced AMPK activity was lower in *Tg-Prkag3*^{225Q} and *Prkag3*$^{-/-}$ mice (61% and 50%, respectively P<0.05) than with wild-type mice. Under fasted conditions, basal AMPK activity was similar between wild-type and *Tg-Prkag3*^{225Q} mice but blunted (52%; P<0.05) in *Prkag3*$^{-/-}$ mice. In contrast to fed conditions, in EDL muscle from fasted mice, contraction-mediated AMPK activity was similar between genotypes.

Glycogen content

Glycogen content in EDL muscle was assessed at rest and after in vitro contraction, under fed and fasted conditions (**Table 2**). In vitro contraction reduced glycogen content ~50% (P<0.05) in EDL muscle, regardless of genotypes, under either nutritional state. Moreover, in *Tg-Prkag3*^{225Q} mice, glycogen content was higher at rest (>60%; P<0.05) and after in vitro contraction (>50%, P<0.05) compared with wild-type and *Prkag3*$^{-/-}$ mice regardless of nutritional state. In fed and fasted *Prkag3*$^{-/-}$ mice, glycogen content was ~35% (P<0.05) lower at rest compared with wild-type mice. Regardless of nutritional state, however, glycogen content in EDL muscle from *Prkag3*$^{-/-}$ mice after contraction was similar to that of wild-type mice.

Relationship between AMPK activity and glycogen content

In wild-type and *Tg-Prkag3*^{225Q} mice, an inverse correlation between AMPK activity and glycogen content was observed (**Fig. 1**, R^2= −0.63 and −0.64, respectively). This supports the concept that glycogen has a feedback regulation on AMPK activity. In *Prkag3*$^{-/-}$ mice, however, glycogen content and AMPK activity were unrelated (R^2= −0.36), indicating that expression of the AMPKγ3 isoform is necessary for glycogen feedback on AMPK.

Skeletal muscle fatigue

Resistance to skeletal muscle fatigue was determined in EDL muscle after electrical stimulation. Fatigue was measured as the time elapsed until 50% of the peak force was generated (**Fig. 2*A***). EDL muscles from *Tg-Prkag3*^{225Q} mice were fatigue resistant, as demonstrated by an increased ability to sustain work compared with wild-type mice (difference=+0.21±0.06 min, P<0.002 in an ANOVA analysis combining data from fed and fasted mice). In contrast, EDL muscle from *Prkag3*$^{-/-}$ mice were fatigue prone and had a decreased time to fatigue (difference=−0.17±0.07 min; P<0.02 vs. wild-type mice in an ANOVA analysis combining data from fed and fasted mice). Time to fatigue was similar in *Prkag3*$^{-/-}$ and wild-type mice under fasting conditions. Skeletal muscle fatigue was positively correlated with glycogen content (Fig. 2*B*, R^2=0.79).

TABLE 2. *Glycogen content in skeletal muscle*[a]

	Fed		Fasted	
	Basal	Contraction	Basal	Contraction
Wild-type	15.68 ± 1.17	6.57 ± 0.73†	7.38 ± 0.64	3.33 ± 0.22†
Tg-Prkag3^{225Q}	28.67 ± 3.34*	14.16 ± 1.18†*	11.76 ± 0.75*	5.01 ± 0.63†*
Prkag3$^{-/-}$	10.29 ± 0.83*	4.75 ± 0.32†	4.92 ± 0.61*	3.35 ± 0.16

[a] Skeletal muscle glycogen content was measured in EDL muscle from fed or fasted *wild-type, Tg-Prkag3*^{225Q}, and *Prkag3*$^{-/-}$ mice before or after *in vitro* muscle contraction. Values are expressed as mmol × kg^{-1} (n=4–10). * $P < 0.05$ vs. *wild-type* of same condition. † $P < 0.05$ vs. basal of same genotype.

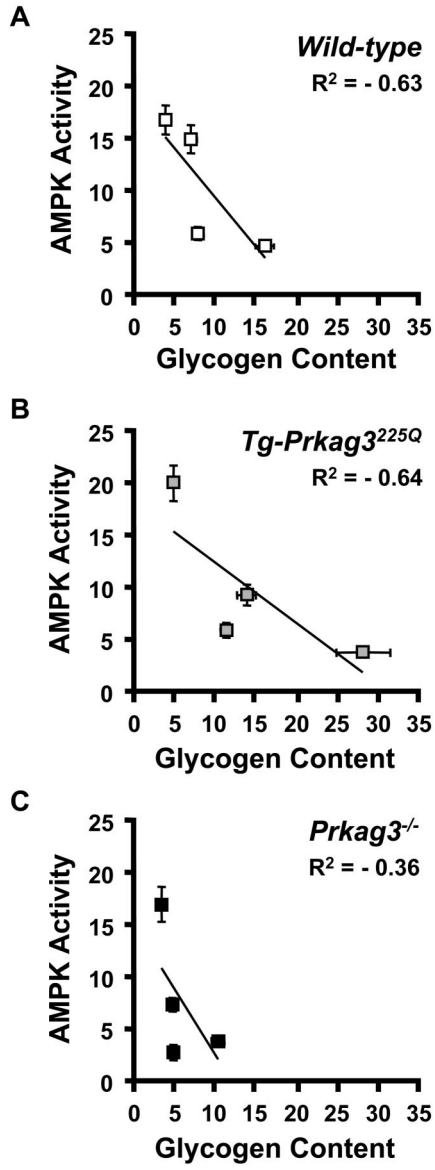


Figure 1. Relationship between glycogen content and AMPK activity in rested and contracted skeletal muscle from fed or fasted *A)* wild-type (*wt*, open bar), *B)* AMPKγ3 mutant-overexpressing ($g3^{225Q}$, shaded bar), and *C)* AMPKγ3 knockout ($g3^{-/-}$, filled bar) mice. Data represent means ± SE. R^2 values are located on each graph.

Glucose oxidation

Glucose oxidation was determined after in vitro contraction of EDL muscles from fasted mice (**Fig. 3**). In *Tg-Prkag3*^{225Q} mice, glucose oxidation was reduced 45% ($P<0.05$) after in vitro contraction. Conversely, glucose oxidation was increased 44% ($P<0.05$) in *Prkag3*$^{-/-}$ mice after in vitro contraction.

DISCUSSION

The primary goal of this investigation was to elucidate the role of AMPK in the regulation of glycogen content

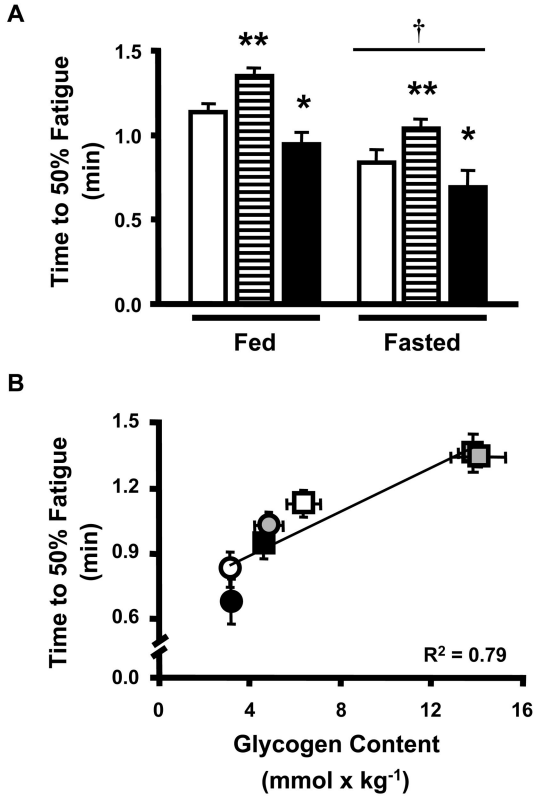


Figure 2. Electrically stimulated muscle fatigue and relationship to glycogen content in skeletal muscle from fed and fasted mice. *A)* Wild-type (*wt*, open bar), AMPKγ3 mutant-overexpressing ($g3^{225Q}$, vertical lines), and AMPKγ3 knockout ($g3^{-/-}$, filled bar) mice. Isolated EDL muscles were electrically stimulated and time to fatigue was recorded. Values are time elapsed before 50% of peak force (Fp) was generated (n=5–16). *B)* Glycogen content correlated with time to 50% fatigue in contracted skeletal muscle from fed (square) and fasted (circle) mice. Data are means ± SE. R^2 value is located on graph. $*P < 0.05$ or $**P < 0.01$ vs. wild-type of same condition. $\dagger P < 0.05$ vs. fed conditions in the same genotype.

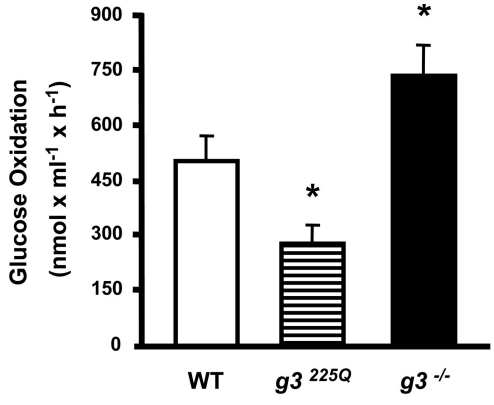


Figure 3. Glucose oxidation after electrical stimulation in EDL skeletal muscle from fasted mice. Wild-type (*wt*, open bar), AMPKγ3 mutant-overexpressing ($g3^{225Q}$, vertical lines), and AMPKγ3 knockout ($g3^{-/-}$, filled bar) mice. Data are means ± SE. Results are nmol × mL^{-1} × h^{-1} (n=5–9). $*P < 0.05$ vs. *wild-type*.

and skeletal muscle ergogenics using *Tg-Prkag3*^{225Q} and *Prkag3*$^{-/-}$ mice. We provide direct evidence that AMPK is required for glycogen supercompensation and exercise performance in skeletal muscle. The inverse changes in glucose metabolism between the genotypes favor glucose incorporation to glycogen in the *Tg-Prkag3*^{225Q} mice and a greater reliance on glucose oxidation in *Prkag3*$^{-/-}$ mice. Thus, glucose handling and ergogenics in response to intense anaerobic exercise are under intrinsic control of the AMPKγ3 subunit.

We have previously reported that heterotrimeric complexes containing the AMPKγ3 R225Q mutation result in an increase in AMPK activity and phosphorylation on Thr^{172} in an AMP-independent manner and diminished AMP dependence in transfected Cos cells (13). Thus, the R225Q mutation is a loss-of-function mutation that abolishes allosteric regulation by AMP/ATP, which is predicted to increase basal AMPK activity. Nevertheless, in rested muscle from *Tg-Prkag3*^{225Q} mice AMPK activity was unaltered. In fed AMPK R225Q mutant pigs, AMPK activity was reduced (16). While these results are paradoxical, the excessive skeletal muscle glycogen content in *Tg-Prkag3*^{225Q} mice and AMPK R225Q mutant pigs may feed back and inhibit AMPK activity. Several lines of evidence suggest high glycogen content inhibits AMPK activity in skeletal muscle (11, 22, 23). Therefore, a plausible explanation for the lack of a further elevation in AMPK activation in *Tg-Prkag3*^{225Q} vs. wild-type mice can be attributed to the 70% higher glycogen content. Although we are unable to biochemically distinguish the AMPK associated with different gamma isoforms, when muscle glycogen levels are depleted in *Tg-Prkag3*^{225Q} mice, contraction-induced AMPK activity is increased. In contrast, the reduction in AMPK activity after contraction in fed *Prkag3*$^{-/-}$ mice is unrelated to glycogen content and suggests that the γ3 subunit is responsible for the major proportion of contraction-induced AMPK activation. However, under fasted conditions, contraction-induced AMPK activity is unaltered in *Prkag3*$^{-/-}$ mice, possibly a consequence of a more general AMPK activation in the energy-deprived fasted muscle. Moreover, glycogen content is negatively correlated with AMPK activity in wild-type and *Tg-Prkag3*^{225Q} mice, but not in *Prkag3*$^{-/-}$ mice (Fig. 1), highlighting the potential importance of the γ3 isoform in glycogen feedback on AMPK.

Several lines of evidence indicate AMPK is a glycogen synthase kinase; thus, active AMPK would be predicted to prevent glycogenesis in skeletal muscle. AMPK phosphorylates and inactivates muscle glycogen synthase in cell-free assays at site 2 (24). In vivo, AMPK is a glycogen synthase kinase in skeletal muscle in response to AICAR stimulation (23) and exercise (25), phosphorylating site 2 and leading to inactivation of glycogen synthase. AMPK activity is negatively correlated with glycogen synthase activity during exercise (7); thus, active AMPK should prevent glycogenesis in skeletal muscle. In vivo, however, the effect of AMPK activation on glycogen synthase may also be indirectly regulated through the action of other glycogen synthase phosphatases or kinases (25). AICAR and exercise activate AMPK and inactivate glycogen synthase (23, 25) while simultaneously activating glucose transport (26, 27). The effects of AICAR or exercise on glucose transport may predominate and facilitate, rather than retard, glycogenesis through increased cellular levels of glucose-6-phosphate, which may override any inhibitory phosphorylation of glycogen synthase by allosteric activation. In support of this hypothesis, treatment of animals with AICAR actually increases skeletal muscle glycogen content (27, 28). Moreover, in transgenic mice overexpressing a kinase-dead AMPKα2 subunit (α2KD) and AMPKα2 knockout mice, AMPK activity is severely blunted, and glycogen levels (14, 17) and rates of resynthesis after exercise (29) are markedly impaired. Similarly, skeletal muscle glycogen levels (Table 2) and rates of resynthesis after exercise are blunted in *Prkag3*$^{-/-}$ mice (13).

AMPK has been linked to exercise tolerance and performance (14, 29). AMPKα2KD mice have decreased spontaneous physical activity as determined by volunteer wheel running and increased muscle fatigue in response to repeated electrical stimulation of isolated muscle (14, 29). Here we report modification of the AMPKγ3 subunit influences muscle ergogenics. Work performance is enhanced in skeletal muscle from *Tg-Prkag3*^{225Q} mice under fed and fasted conditions. In contrast, skeletal muscle from *Prkag3*$^{-/-}$ mice is fatigue prone during exercise. Based on in vitro observations in isolated skeletal muscle, we provide evidence that the AMPKγ3 isoform enhances muscle ergogenics by altering glycogen levels and that this effect is independent of capillary density or serum factors. Moreover, the enhanced work performance in *Tg-Prkag3*^{225Q} mice is independent of fiber type changes (13). It is interesting that work performance in *Tg-Prkag3*^{225Q}, *Prkag3*$^{-/-}$, and wild-type mice was positively correlated to the level of skeletal muscle glycogen. Thus, the increase in glycogen content, rather than AMPK per se, is a major determinant of anaerobic muscle performance.

To determine the role of the AMPKγ3 isoform on glycogenesis after anaerobic exercise, we measured glycogen resynthesis rates at 2.5 and 6 h of recovery after in vitro contraction (data not shown). However, glycogen resynthesis was minimal under these conditions. Since the early phase glycogen resynthesis is slow after eccentric and high-intensity resistance exercise (30, 31), the minimal glycogen synthesis rate after in vitro contraction would be expected. After moderate exercise, such as swimming in *Tg-Prkag3*^{225Q} mice (13) or treadmill running in AMPK R200Q pigs (16), the rate of glycogen resynthesis is rapidly increased than wild-type animals. In contrast, α2KD mice and *Prkag3*$^{-/-}$ mice have lower resting glycogen content and defects in glycogen resynthesis after exercise compared with wild-type mice (14, 29). Together, these findings support the functional role of AMPK and

highlight γ3 containing complexes in mediating glycogenesis after exercise.

To investigate glucose handling after glycogen-depleting contraction, we used indirect methods to determine the potential for glycogen resynthesis. Once glucose is transported across the plasma membrane, the glycolytic fate of glucose is oxidation or glycogen synthesis. We have previously reported that glucose uptake after electrically stimulated muscle contraction is similar among *Tg-Prkag3*^{225Q}, *Prkag3*$^{-/-}$, and wild-type mice (13). This would suggest that alterations in glucose handling partly contribute to the changes in glycogen content in genetically modified AMPKγ3 mice. Indeed, glucose oxidation was markedly reduced in *Tg-Prkag3*^{225Q} mice and increased in *Prkag3*$^{-/-}$ mice. This observation is further supported by the finding that *Tg-Prkag3*^{225Q} and *Prkag3*$^{-/-}$ mice have a reciprocal rate of glycogen resynthesis after an endurance bout of exercise (13).

In conclusion, the *Prkag3*^{225Q} mutation increases skeletal muscle ergogenics by enhancing glycogen content. Moreover, in response to anaerobic exercise, the *Prkag3*^{225Q} mutation suppresses glucose oxidation and facilitates glycogenesis. The AMPKγ3 isoform has intrinsic control over anaerobic exercise responses, since ablation of the isoform influences glucose metabolism and muscle ergogenics. Functional studies of the AMPKγ3 isoform further support the close connection between glycogen content and exercise performance in skeletal muscle. FJ

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